# ARTIFICIAL INTELLIGENCE

## SECTION 1: AI FUNDAMENTALS & HISTORY

### 1.1 Core Definitions

• **Artificial Intelligence (AI):** A branch of science dealing with helping machines find solutions to complex problems in a human-like fashion. It enables machines to function with or without human supervision.

• **The Agent:** An entity that perceives its environment through **sensors** (e.g., cameras, microphones) and acts upon it using **actuators/effectors** (e.g., motors, screens, speakers).

• **The Environment:** The surrounding world in which the agent operates (e.g., a room for a robot, a road for a car).

• **Cognitive Skills:** AI programming focuses on Learning, Reasoning, Problem Solving, Perception, and Linguistic Intelligence.

### 1.2 Timeline of AI (History)

• **1941:** First electronic computer.

• **1956: John McCarthy** coined the term "Artificial Intelligence" at the first AI conference.

• **1969: Shakey** (first general-purpose mobile robot) built; it acted with purpose rather than just following a list.

• **1997:** IBM's **Deep Blue** defeated world chess champion Garry Kasparov.

• **2002:** First successful robotic vacuum cleaner (Roomba).

• **2020 (COVID-19):** Baidu's **LinearFold AI** predicted the SARS-CoV-2 RNA sequence in just 27 seconds (120x faster than traditional methods).

• **2020-2024:** Rise of **Generative AI** (ChatGPT, DALL-E, Google Bard/Gemini) creating original content.

### 1.3 The 4 Types of AI (Based on Capability)

1. **Purely Reactive:**
   ◦ No memory of the past. Operates only on current data.
   ◦ *Examples:* IBM Deep Blue, Google AlphaGo.
2. **Limited Memory:**
   ◦ Uses historical data to make decisions. Most modern AI falls here.
   ◦ *Examples:* Self-driving cars (observing traffic speed), Chatbots (remembering recent context).
3. **Theory of Mind:**
   ◦ Understands human emotions, beliefs, and social interaction. (Concept stage; does not exist yet).
   ◦ *Goal:* Socially interacting robots (e.g., robots in healthcare).
4. **Self-Awareness:**
   ◦ The machine has its own consciousness and desires. (Hypothetical/Ultimate goal).

## SECTION 2: KEY AI DOMAINS

### 2.1 Data Science

• **Definition:** Interdisciplinary field using algorithms to extract insights from structured and unstructured data.

• **Goal:** Make decisions or predict outcomes (e.g., Price prediction, Recommendation systems).

• **Tools:** Data mining, Machine Learning, Big Data.

### 2.2 Computer Vision (CV)

• **Definition:** Enabling machines to interpret and decide based on visual input (images/video).

• **Applications:**
   ◦ **Facial Recognition:** Unlocking phones, DeepFace (Facebook tagging).
   ◦ **Healthcare:** Detecting tumors in brain scans (Semantic Segmentation).
   ◦ **Object Detection:** Google Lens identifying flowers; Self-driving cars identifying pedestrians vs. stop signs.

### 2.3 Natural Language Processing (NLP)

• **Definition:** Enabling computers to understand, interpret, and generate human language.

• **Two Components:**
   1. **NLU (Understanding):** Reading and interpreting meaning.
   2. **NLG (Generation):** Producing text/response.

• **The 5 Stages of NLP Analysis:**
   1. *Lexical:* Breaking text into words/paragraphs.
   2. *Syntactic:* Grammar checking.
   3. *Semantic:* Extracting dictionary meaning.
   4. *Disclosure Integration:* Context from preceding sentences.
   5. *Pragmatic:* Deriving real-world meaning/re-interpretation.

• **Applications:** Spam filtering (Gmail), Sentiment Analysis (Social media monitoring), Plagiarism checkers, Virtual Assistants (Siri/Alexa).

## SECTION 3: MACHINE LEARNING (ML)

**Definition:** A subset of AI where computers learn from data without being explicitly programmed.

3.1 Supervised Learning (Teacher Present)

- **Mechanism:** Input ($x$) and Output ($Y$) are known. The algorithm maps $Y=f(x)$.
- **Goal:** Train on labeled data to predict outcomes for new data.
- **Categories:**
  - **Regression:** Predicting **continuous values** (e.g., Salary based on experience, House prices, Weather, Stock market).
    - *Algorithms:* Linear Regression.
  - **Classification:** Predicting **discrete labels/categories** (e.g., Spam vs. Ham, Yes vs. No for loan approval).
    - *Algorithms:* Logistic Regression, Decision Tree, Random Forest, SVM, Naïve Bayes.

3.2 Unsupervised Learning (No Teacher)

- **Mechanism:** Only Input ($X$) is given. No output labels.
- **Goal:** Find hidden patterns, structures, or groups in data.
- **Categories:**
  - **Clustering:** Grouping similar items (e.g., Customer segmentation by spending behavior).
    - *Algorithms:* K-Means, Hierarchical Clustering.
  - **Association:** Finding rules/relationships (e.g., "People who buy Bread also buy Butter").
    - *Application:* **Market Basket Analysis** (e.g., D-Mart store layout).
    - *Algorithms:* Apriori, Eclat.

3.3 Reinforcement Learning (Experience Based)

- **Mechanism:** Agent learns through trial and error using a **Reward/Penalty** system.
- **Goal:** Maximize reward in a specific situation.
- **Applications:** Robotics, Traffic light control, Game playing (Chess/Go).
- **Algorithms:** Q-Learning, Deep Q-Learning.

3.4 The ML Process Flow

- **Data Split:** Typically **80% Training** data (to teach the model) and **20% Testing** data (to verify accuracy).
- **Workflow:** Data → Training → ML Algorithm → Trained Model → Prediction.

## SECTION 4: DEEP LEARNING & NEURAL NETWORKS

4.1 ML vs. Deep Learning (DL)

| Feature | Machine Learning | Deep Learning |
|---|---|---|
| **Data Size** | Small / Structured | Huge / Unstructured |
| **Feature Extraction** | Manual (Human selects features) | Automated (Network learns features) |
| **Performance** | Good on simpler tasks | Superior on complex tasks (Images/NLP) |
| **Hardware** | Lower requirements | High-end (GPUs) required |

4.2 Artificial Neural Networks (ANN)

Inspired by the human brain.

- **Biological Neuron:** Dendrites (Input), Soma (Processing), Axon (Output).
- **Artificial Neuron (Perceptron):**
  - **Structure:** Input Layer → Hidden Layers (Processing) → Output Layer.
  - **Math Formula:** $y=ActivationFunction(\sum(Weight \times Input)+Bias)$.
    - **Weights ($w$):** Determine the influence of an input.
    - **Bias ($b$):** Allows the model to shift; improves flexibility.
  - **XOR Problem:** A single-layer perceptron **cannot** solve non-linear problems like XOR. You need a **Multi-layer Perceptron (MLP)** for that.

4.3 Activation Functions

Functions that decide if a neuron should "fire" (output a signal). They introduce **non-linearity**.

- **Sigmoid:** Outputs values between 0 and 1. Used for probability/binary classification.
- **ReLU (Rectified Linear Unit):** Outputs 0 if input is negative, or the input itself if positive. Most common in hidden layers.

4.4 Types of Neural Networks

1. **ANN:** Standard Feed-Forward network.
2. **CNN (Convolutional Neural Network):** Specialized for **Grid data (Images/Video)**. Uses "Pooling" and "Convolution" layers for feature extraction.
3. **RNN (Recurrent Neural Network):** Specialized for **Sequential data (Time Series, Text, Audio)**. Has loops allowing information to persist.

## SECTION 5: ADVANCED & TRENDING CONCEPTS

5.1 Fuzzy Logic

- **Concept:** Handles uncertainty. Instead of Boolean logic (0 or 1 / True or False), it uses **degrees of truth** between 0 and 1 (e.g., "Very hot," "Slightly hot," "Cold").
- **Key Term: Membership Function** (defines how input maps to a degree of truth between 0 and 1).
- **Applications:** Temperature control (ACs), Washing machines.

5.2 Genetic Algorithms (GA)

- **Concept:** Optimization algorithms inspired by **natural evolution** and survival of the fittest.
- **Key Operators:**
  - **Selection:** Choosing the best solutions.
  - **Crossover (Reproduction):** Combining parents to create offspring.
  - **Mutation:** Random changes to maintain diversity.
- **Population:** Refers to the set of potential solutions.

5.3 Modern Trends

- **Kavya Mehra:** India's first AI-driven mom influencer, powered by data from real moms.
- **Generative AI:** Focuses on creating new content (Text, Voice, Images, Code).

**SECTION 6: TECHNICAL BASICS (PYTHON & MATH)**

6.1 Python Data Types

- **Mutable (Can be changed):** List, Dictionary, Set.
- **Immutable (Cannot be changed):** Tuple, String, Integer, Numeric.

6.2 Arithmetic Operators & Precedence

- **Precedence Order (Highest to Lowest):**
  1. ** (Exponentiation)
  2. *, /, //, %
  3. +, -
- **Examples from Source:**
  - 2 * 2 ** 10 → 2 * 1024 = **2048**.
  - 3 * 2 ** 10 → 3 * 1024 = **3072**.
  - // (Floor Division): Divides and rounds down to the nearest whole number (e.g., 32 // 3 = 10).

**<u>AI QUIZ - 1</u>**

**1. Which system in AI is used by understand and converse with humans through voice command?**

- Siri
- Alexa
- Cortana
- All above

**2. Which of the following is an example of a unsupervised learning problem?**

- Population growth prediction
- Market forecasting
- Market basket analysis
- Stock prediction

**3. What is the difference between supervised and unsupervised learning?**

- Supervised learning requires labelled data while unsupervised learning does not.
- Unsupervised learning requires labelled data while supervised learning does not.
- Supervised learning does not require data while unsupervised learning does.
- There is no difference between supervised and unsupervised learning.

**4. Artificial Intelligence is about _____.**

- Programming on Machine with your Own Intelligence
- Playing a game on Computer
- Putting your intelligence in Machine
- Making a machine Intelligent

**5. The term "artificial intelligence" was coined by _____.**

- Yann LeCun
- John McCarthy
- Seyedali Mirjalili
- Dr Pushpak Bhattacharya

**6. What is the definition of data science?**

- The study of collecting and analysing data
- The study of designing databases
- The study of computer hardware
- The study of data encryption techniques

**7. Machine learning is a subset of which of the following.**

- Deep learning
- Data learning
- Artificial Intelligence
- Advance computing

**8. Which are types of Machine learning?** I-Semi-structured learning II-Supervised learning III-Structured learning IV-Unsupervised learning V-Reinforcement learning

- I, III, V
- II, IV, V
- II, III
- I, V

**9. Which of the following are the application of AI?** I-Data Security II-Gaming III-Social Media IV-Automotive Industry V-Robotics

- I, III, V
- II, IV
- I, III, IV, V
- I, II, III, IV, V

**10. What is machine translation?**

- Converting a human language to another
- Converting a human language to machine language
- Converting any human language to English
- Converting Opcode to machine language

**<u>AI QUIZ – 2</u>**

**Question 1** Which of the following is an examples of supervised learning?

- SVM (Support Vector Machine)
- PCA (Principal Component Analysis)
- apriori algorithm
- K mean clustering

**Question 2** RNN Stands for _____.

- Recurring Neural Network
- Recursive Neural Network
- Removable Neural Network
- Recurrent Neural Network

**Question 3** Which of the following is not the goal of exploratory data analysis?

- To identifying the cause.
- To identify pattern in data
- To identify relationship in data
- To do unbiased evaluation

**Question 4** Which is true for neural networks? **I** - It has set of nodes and connections **II** - Each node computes it's weighted input **III** - It is neurally implemented mathematical model

- III, II
- I
- I, II
- I, II, III

**Question 7** Which algo is used for text classification?

- K means clustering
- Naïve bayes
- PCA
- Linear regression

**Question 8** Which of the following is not correct in case of advantage of AI?

- It's fast
- It reduces human error
- Implementation of it is cheaper
- High availability

**Question 9** The main and most important feature of RNN is _____.

- Visible State
- Hidden state
- Present State
- developed State

**Question 10** Data can be visualized using? **I** - graphs **II** - rings **III** - charts **IV** - maps

- I, II, III
- II, III, IV
- I, III, IV
- I, II, III, IV

**AI QUIZ – 3**

**Question 1** Artificial neural network is used for: I. Classification II. Clustering III. Pattern recognition IV. Robotics

- A) I, III, IV
- B) I, II
- C) II, III, IV
- D) I, II, III, IV

**Question 2** CNN stands for _____.

- Convolutional Neutron Networks
- Convolutional Neural Networks
- Convolutional Natural Networks
- Conversion Natural Networks

**Question 3** Which of the following is an example of supervised learning?

- Apriori
- Q-Learning
- naive bayes classification
- PCA

**Question 4** The term _____ refers to different processes, though the most common is putting data into ordered rows and columns in a matrix format (also called flat-file transformation).

- Cleansing
- Inspecting
- Transforming
- Modelling

**Question 5** Which algorithm is commonly used for sentiment analysis?

- Linear regression
- Logistic regression
- K-means clustering
- CNN

**Question 6** The _____ laws allow robots to work with humans without any risk.

- two
- three
- four
- six

**Question 7** Which of the following is not a major type of Machine learning?

- Supervised learning
- Semi-Supervised learning
- Unsupervised learning
- Reinforcement learning

**Question 8** Algorithm is:

- It uses machine-learning techniques. Here program can learn from past experience and adapt themselves to new situations
- Computational procedure that takes some value as input and produces some value as output
- Science of making machines performs tasks that would require intelligence when performed by humans
- None of these

**Question 9** _____, a pioneering American computer scientist and inventor, earned the title "Father of Artificial Intelligence."

- Raj Reddy
- Nathaniel Rochester
- John McCarthy
- Marvin Minsky

**Question 10** Which statistical measure provides a central tendency of a dataset?

- Variance
- Range
- Median
- Standard deviation

**Question 11** What is the purpose of a decision tree algorithm?

- To classify the data in categorise
- To perform clustering analysis
- To optimize computational resources
- To calculate summery statistics

**Question 12** What is the purpose of feature scaling in machine learning?

- To standardize the range of numerical features
- To reduce the dimensionality of the feature space
- To convert categorical features into numerical features
- To introduce new features into the model

**Question 13** Which of the following is the branch of Artificial Intelligence?

- Cyber security
- Network Design
- Machine Learning
- NPL

**Question 14** Which of the following is an example of a dimensionality reduction technique?

- PCA
- SVM
- KNN
- CNN

**Question 15** Bayesian network, which consists of _____ showing how events affect each other.

- roads
- graphs
- trees
- traffic

**Question 16** What is the primary goal of exploratory data analysis?

- To identifying the cause.
- To identify pattern and relationship in data
- To check the unseen data
- To evaluate result

**Question 17** Suppose, your target variable is whether a passenger will have survived or not using Decision Tree. What type of tree do you need to predict the target variable?

- classification tree
- regression tree
- clustering tree
- dimensionality reduction tree

**Question 18** Which of the following is an example of a supervised learning problem?

- Image classification
- Market segmentation
- Fraud detection
- Social network analysis

**Question 19** Decision trees are used to solve: I - classification problems II - regression problems III - to predict continuous outcomes

- I
- II, III
- I, III
- I, II, III



**1.**

**What is the Turing Test used for in AI?**

**A. To test computer memory**
**B. To evaluate programming skills**
**C. To determine whether a machine can imitate human intelligence**
**D. To check internet speed**

**Answer: C**

**2.**

**Which of these is an example of Narrow AI?**

**A. A robot that has emotions**
**B. A machine that solves any human problem**
**C. A language translator app**
**D. A self-aware robot**

**Answer: C**

**3.**

**What domain of AI focuses on enabling machines to interpret and understand visual information?**

**A. Natural Language Processing**
**B. Expert Systems**
**C. Computer Vision**
**D. Speech Recognition**

**Answer: C**

Sunbeam

4.

Which type of AI can perform only specific tasks and lacks consciousness?

A. Artificial General Intelligence
B. Artificial Super Intelligence
C. Artificial Narrow Intelligence
D. Conscious AI

Answer: C

5.

Which domain of AI enables machines to understand and respond to human language?

A. Robotics
B. Natural Language Processing
C. Machine Learning
D. Deep Learning

Answer: B

6.

Which domain of AI would most likely be used in facial recognition systems?

A. Expert Systems
B. Computer Vision
C. NLP
D. Robotics

Answer: B





7.

Deep Learning is a subset of which AI domain?

A. Natural Language Processing
B. Robotics
C. Machine Learning
D. Expert Systems

Answer: C



SunBeam

1.

In which domain of AI is reinforcement learning most commonly used?

A. Expert Systems
B. Robotics
C. Speech Recognition
D. Database Management

Answer: B

2.

Which of the following is not a typical task in Data Science?

A. Data cleaning
B. Data visualization
C. Data encryption
D. Predictive modeling

Answer: C

3.

The process of feeding the system both the input and the correct output during training is called:

A. Unsupervised learning
B. Reinforcement learning
C. Supervised learning
D. Semi-supervised learning

Answer: C





4.

Which of the following is a supervised learning algorithm?

A. K-Means
B. Decision Tree
C. PCA
D. DBSCAN

Answer: B

5.

Which of the following problems is best solved using supervised learning?

A. Grouping customers based on purchasing behavior
B. Predicting the price of a house based on features
C. Detecting anomalies in a dataset without labels
D. Reducing the number of features in a dataset

Answer: B

6.

In unsupervised learning, the model works on:

A. Labeled data
B. Only numerical data
C. Unlabeled data
D. Real-time streaming data only

Answer: C

7.

Linear Regression is used for:

A. Classification
B. Clustering
C. Dimensionality Reduction
D. Prediction of continuous values

Answer: D





1.

What is Artificial Intelligence (AI)?

A. Programming computers to create websites
B. Teaching computers to play video games
C. The simulation of human intelligence in machines
D. Building faster computer processors

Answer: C

2.

Which of the following best represents the goal of AI?

A. Replace all human jobs
B. Mimic and eventually surpass human cognitive abilities
C. Make computers more entertaining
D. Improve physical strength of robots

Answer: B

3.

Which is not a commonly accepted type of AI?

A. Narrow AI
B. General AI
C. Super AI
D. Creative AI

Answer: D



SunBeam

4.

Which of the following is not a main domain of AI?

A. Machine Learning
B. Computer Vision
C. Blockchain
D. Natural Language Processing

Answer: C

5.

When was the term “Artificial Intelligence” first coined?

A. 1920
B. 1956
C. 1972
D. 1984

Answer: B

6.

Which of the following best describes a key feature of AI systems?

A. They never make mistakes
B. They always need internet access
C. They can learn from experience
D. They are only used for gaming

Answer: C





7.

Who is considered one of the founding fathers of Artificial Intelligence?

A. Steve Jobs
B. Alan Turing
C. Mark Zuckerberg
D. Elon Musk

Answer: B

1.

What is the role of Data Science in AI?

A. Designing computer chips
B. Managing human resources
C. Extracting insights & patterns from data to support AI models
D. Building physical robots

Answer: C

2.

What programming languages are most commonly used in Data Science for AI applications?

A. HTML and CSS
B. Python and R
C. JavaScript and PHP
D. Ruby and Swift

Answer: B

3.

What is the main goal of machine learning?

A. To program computers manually for every task
B. To develop algorithms that can learn from and make predictions on data
C. To create a new operating system
D. To improve database indexing

Answer: B





4.

Which of the following is a real-world application of machine learning?

A. Spam email detection
B. Typing on a keyboard
C. Printing documents
D. Playing a DVD

Answer: A

5.

In supervised learning, the model is trained on:

A. Unlabeled data
B. Data with missing values
C. Labeled data
D. Random data

Answer: C

6.

Classification is a type of:

A. Unsupervised learning
B. Supervised learning
C. Reinforcement learning
D. None of the above

Answer: B

7.

Which of the following is an unsupervised learning algorithm?

A. Logistic Regression
B. K-Means Clustering
C. Naive Bayes
D. Random Forest

Answer: B

SunBeam